Simulator

Requirements and Technical Specification

Start Line Fusion Algorithm Simulator

Document Purpose

Develop a browser-based simulation platform used to design, test, tune, and validate a sensor fusion algorithm capable of calculating highly accurate race-start metrics for sailing vessels approaching a start line.

The simulator shall model GPS, UWB, LoRa communications, vessel dynamics, start-line geometry, radio contention, measurement errors, and sensor update rates to evaluate algorithm performance under realistic operating conditions.

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1. System Objectives

The system shall simulate a sailing race start environment and provide the data necessary to develop and optimize an onboard fusion algorithm capable of determining:

1. Distance to Port Mark
2. Distance to Starboard Mark
3. Distance to Start Line given Vessel's current trajectory
4. Time to Reach Start Line
5. Predicted Undershoot Distance
6. Predicted Overshoot Distance
7. Time Early/Late at Starting Signal
8. Confidence Level of Calculated Results

The simulator shall support varying environmental conditions, communication delays, sensor errors, and vessel behaviors.

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2. System Overview

The simulated environment consists of:

Fixed Infrastructure

Port Start Mark

Contains:

- GPS receiver
- UWB Anchor
- UWB Tag
- LoRa radio

Starboard Start Mark

Contains:

- GPS receiver
- UWB Anchor
- UWB Tag
- LoRa radio

The marks shall continuously:

- Determine their own GPS position
- Measure distance to the opposite mark using UWB
- Broadcast status information via LoRa

Vessel

Each vessel contains:

- GPS receiver
- UWB Anchor
- LoRa radio

The vessel shall:

- Determine its own GPS position

- Determine speed over ground (SOG)
- Determine course over ground (COG)
- Receive LoRa broadcasts
- Participate in UWB ranging operations
- Execute the fusion algorithm

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3. Start Line Geometry

The start line is defined by:

Port Mark ---------------- Starboard Mark

The line exists between the actual positions of the two marks.

The simulator shall maintain:

- True mark positions
- GPS-derived mark positions
- UWB-derived mark separation

The system shall support:

- Arbitrary line lengths
- Line movement
- GPS drift
- Mark movement due to current or wave action

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4. Available Measurements

GPS Measurements

Boat GPS

Provides:

- Latitude
- Longitude
- Timestamp

- Speed Over Ground
- Course Over Ground

Characteristics:

Parameter	Typical Accuracy
Position	1-5 meters
Speed	±0.05 knots
Heading from movement	Excellent above 2 knots

Simulator shall allow configurable error models.

Mark GPS

Provides:

- Latitude
- Longitude
- Timestamp

Characteristics:

Parameter	Typical Accuracy
Position	1-5 meters

UWB Measurements

Boat-to-Port Distance

Provides:

D_BP

Distance between Boat and Port Mark.

Expected Accuracy:

±0.1m

Configurable in simulator.

Boat-to-Starboard Distance

Provides:

D_BS

Distance between Boat and Starboard Mark.

Expected Accuracy:

±0.1m

Configurable.

Port-to-Starboard Distance

Measured by:

- Port Anchor → Starboard Tag
- Starboard Anchor → Port Tag

Produces:

D_PS
D_SP

These measurements may differ slightly.

Fusion algorithm shall support:

D_LINE = (D_PS + D_SP) / 2

or alternative estimation methods.

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5. Communications Model

LoRa Network

LoRa shall be modeled as:

- Shared medium
- Broadcast only
- Low bandwidth
- Potential packet loss
- Variable latency

Simulator shall allow configuration of:

Parameter	Range
Packet loss	0-50%
Latency	0-10 sec
Throughput	configurable
Update interval	configurable

Broadcast Messages

Mark Status Message

Contains:

Mark ID
GPS Position
Timestamp
D_PS
D_SP
Battery
Health Status

Boat Status Message

Contains:

Boat ID
GPS Position
Speed
Course
Timestamp

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6. UWB Access Management

Problem

Multiple boats cannot continuously range simultaneously.

Access to UWB resources must be prioritized.

Priority Rules

Priority shall be based upon proximity to the start line.

Closest boats receive highest ranging frequency.

Example:

Distance to Line	UWB Rate
< 25m	10 Hz
25-50m	5 Hz
50-100m	2 Hz
>100m	1 Hz

The simulator shall allow configurable scheduling algorithms.

Scheduling Algorithms

Support:

Fixed Priority

Closest boats first.

Dynamic Priority

Priority score:

P =
Distance Weight +
Closing Velocity Weight +
Time To Start Weight

TDMA

Time-slotted ranging.

Round Robin

Fair allocation.

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7. Fusion Algorithm Inputs

The simulator shall expose all raw measurements.

Inputs include:

Boat GPS Position

Boat Speed

Boat Course

Boat Timestamp

Port GPS Position

Starboard GPS Position

Boat-Port UWB Distance

Boat-Starboard UWB Distance

Port-Starboard UWB Distance

Starboard-Port UWB Distance

LoRa Latency

LoRa Packet Loss

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8. Fusion Algorithm Outputs

The simulator shall evaluate the algorithm’s ability to calculate:

Position Relative to Start Line

Cross Track Distance

Distance to Port Mark

D_BP

Distance to Starboard Mark

D_BS

Time to Line

TTL

Computed using:

DistanceToLine
/
ClosingVelocity

Undershoot

Predicted distance short of line at start signal.

Overshoot

Predicted distance beyond line at start signal.

OCS Prediction

Whether vessel will be over the line at start.

Outputs:

SAFE

MARGINAL

OCS

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9. Truth Model

The simulator shall maintain a hidden ground-truth model.

Ground truth shall include:

- Exact boat position
- Exact mark positions
- Exact speed
- Exact course

Ground truth shall never be visible to the fusion algorithm.

It shall be used only for scoring.

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10. Performance Metrics

The simulator shall calculate:

Distance Error

Estimated - Actual

Time Error

Estimated TTL - Actual TTL

OCS Prediction Accuracy

Percentage correct.

Undershoot/Overshoot Error

Difference between predicted and actual crossing location.

Sensor Contribution Analysis

Impact of:

- GPS only
- UWB only
- GPS + UWB
- GPS + UWB + LoRa

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11. Environmental Simulation

The simulator shall support:

GPS Error

- Random noise
- Multipath
- Drift

UWB Error

- Random noise
- Missed ranges
- NLOS degradation

LoRa Error

- Packet loss
- Delays
- Congestion

Boat Dynamics

- Constant speed
- Acceleration
- Deceleration
- Tacking
- Bearing changes

Start Line Dynamics

- Moving marks
- Current
- Wind-driven drift

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12. Visualization Requirements

The web application shall display:

Map View

- Port Mark
- Starboard Mark
- Start Line
- Boats
- Predicted trajectories

Sensor View

- Raw GPS data
- Raw UWB data
- Raw LoRa messages

Fusion View

- Estimated position
- Estimated line location

- Confidence ellipses

Metrics Dashboard

- Distance errors
- Time errors
- OCS prediction accuracy
- Sensor update rates

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13. Technical Requirements

Front End

Recommended:

- Angular
- TypeScript
- OpenLayers

Simulation Engine

Recommended:

- TypeScript
- RxJS
- Deterministic event scheduler

Architecture

```
Scenario Engine
   │
   ▼
Truth Model
   │
   ▼
Sensor Models
```

(GPS/UWB/LoRa)
|
▼
Fusion Algorithm
|
▼
Scoring Engine
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▼
Visualization Layer

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14. Future Expansion

The simulator should be designed to support:

- 1 to 200 boats
- Additional sensors (IMU, compass, wind)
- RTK GPS
- Multiple start lines
- Fleet-wide optimization
- Replay of real race logs
- Hardware-in-the-loop testing using actual UWB and LoRa devices
- Machine-learning assisted fusion tuning
- Monte Carlo analysis across thousands of race-start scenarios

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Primary Design Goal

The primary purpose of the simulator is not visualization but the development and validation of a fusion algorithm capable of achieving sub-meter accuracy in determining a vessel's position relative to the start line and accurately predicting start timing, undershoot, overshoot, and OCS conditions under realistic racing and communications constraints.

Set it up as a **scenario runner + control panel + results dashboard**.

Best Structure

1. Parameter Control Panel

Let you configure:

Start line

- Port/starboard GPS positions
- True line length
- UWB-measured line length
- Mark drift rate
- Line bias / angle

Boat

- Starting position
- Speed
- Course
- Acceleration
- Heading changes
- Number of boats
- Distance from line priority threshold

Sensors

- GPS noise
- GPS drift
- GPS update rate
- UWB noise
- UWB dropout rate
- UWB update rate
- LoRa packet loss
- LoRa latency
- LoRa update rate

Algorithm

- GPS weighting
- UWB weighting
- LoRa freshness timeout
- smoothing factor
- Kalman/EKF parameters
- confidence thresholds
- UWB priority strategy

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2. Scenario Types

Use repeatable scenarios so you can compare algorithm changes fairly.

Examples:

- Straight-line approach
- Fast approach
- Slow approach
- Reaching start
- Port-end approach
- Starboard-end approach
- Boat accelerating toward line
- Boat bearing away
- Mark drift
- GPS multipath
- UWB packet loss
- LoRa congestion
- 20-boat fleet contention
- 100-boat fleet contention

Each scenario should have a fixed random seed so results are reproducible.

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3. Visualization Layout

Use a three-panel layout.

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──┐
│                     │ Parameter Controls       │
│ Map / Race Geometry        │                   │
│                     │ Sensor + Algorithm Settings │
├──────────────────────────────┤                   │
│ Time-Series Charts         │ Scenario Controls        │
└──────────────────────────────┴──────────────────────────
──┘
```

Key Visualizations

1. Map View

Show:

- True boat position
- GPS-reported position
- Fusion-estimated position
- Port mark
- Starboard mark
- True start line
- Estimated start line
- Boat trajectory
- Projected crossing point

Use different symbols or line styles for:

True position

GPS position

UWB-derived position

Fusion estimate

Predicted path

This is the most important visual.

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2. Distance-to-Line Chart

Plot over time:

True distance to line

Estimated distance to line

Error

This tells you whether the fusion algorithm is useful.

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3. Time-to-Line Chart

Plot:

True time to line

Estimated time to line

Error

This is probably your most important racing metric.

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4. Overshoot / Undershoot Chart

At the start signal, show:

Actual position at start

Predicted position at start

Overshoot/undershoot error

Useful display:

Predicted: 0.8 m short

Actual: 0.2 m over

Error: 1.0 m

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5. Confidence View

Show confidence as:

- Ellipse around estimated boat position
- Error band on distance-to-line chart
- Numeric confidence score

Example:

Distance to line: 12.4 m ± 0.6 m

Time to line: 8.2 s ± 0.5 s

OCS risk: 72%

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6. Sensor Health Timeline

Show when data arrives:

GPS: | | | | | | | |

UWB: | | x | |

LoRa: | | |

Where x means dropped/missing.

This will quickly show whether bad results are algorithmic or caused by missing data.

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7. Error Summary Dashboard

For each run, calculate:

Metric	Value
Mean distance-to-line error	0.42 m
95th percentile error	1.1 m
Max error	2.8 m

Mean time-to-line error	0.31 s
OCS prediction accuracy	96%
False OCS warnings	3
Missed OCS events	1
UWB availability	82%
LoRa packet delivery	71%

Best Workflow

Use this loop:

Create scenario

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Run baseline GPS-only algorithm

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Run UWB-only algorithm

↓

Run GPS + UWB fusion

↓

Run GPS + UWB + LoRa freshness rules

↓

Compare results

↓

Tune algorithm parameters

↓

Run Monte Carlo test

↓

Export scorecard

Most Important Feature

Add an **algorithm comparison mode**.

For the same scenario, run multiple algorithms side-by-side:

Algorithm A: GPS only

Algorithm B: UWB trilateration

Algorithm C: EKF fusion

Algorithm D: EKF + UWB priority scheduling

Then compare:

Distance error

Time-to-line error

OCS prediction accuracy

Undershoot/overshoot error

That will make the simulator genuinely useful rather than just visually impressive.